GREEN STEEL PASSPORT

AI-assisted auditable carbon traceability for scrap-to-steel production

DOMAIN	Clean & Green Technology — Sustainable Manufacturing
PS CATEGORY	Software (Computer Vision + ML Regression + Blockchain)
CORE INNOVATION	Scrap-grading CV + Carbon-intensity regression + Blockchain DPP + Zero-Knowledge verification
REGULATORY CONTEXT	EU CBAM in force since Jan 2026 · India's EAF/IF sector ≈ 47% of national steel capacity



EVIDENCE-LINKED
PER-HEAT CARBON RECORD

SAHAJ

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AI-Assisted Auditable Carbon Traceability for Scrap-to-Steel

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PROBLEM ADDRESSED

Manual, Inconsistent Scrap Grading

India's secondary steel sector (EAF/IF, ~47% of national capacity) grades scrap manually and inconsistently.

(EAF/IF ~47% of capacity · subjective quality calls)

Highest CBAM Exposure, Lowest Proof

This is the world's most CBAM-exposed steel sector, yet MSMEs can't cheaply prove scrap quality or per-batch carbon intensity.

(EU CBAM fully in force since Jan 2026 · lost green premiums)

Claims Without Evidence Don't Hold Up

A carbon number with no link back to furnace data or scrap imagery is not defensible to an auditor or customer.

(need evidence-linked output, not a bare figure)

OUR APPROACH

GRADE THE SCRAP — detect & classify composition/contamination from intake images.

↓

FUSE WITH FURNACE DATA — join CV output to process logs via Heat/Camera/Truck ID (leakage-safe).

↓

REGRESS CARBON INTENSITY — benchmark 5 models under grouped CV; keep intervals, not one number.

↓

EXPLAIN EVERY PREDICTION — SHAP attributes each heat's estimate to specific features.

↓

ISSUE AN AUDITABLE CERTIFICATE — every field tagged measured / CV- / model-derived / provenance.

↓

ANCHOR & VERIFY ON-CHAIN — hash to Hyperledger Fabric; prove claims via zero-knowledge.

PROPOSED SOLUTION

Detect, Then Verify Before Trusting the Grade

A scrap crop is only classified once a calibrated conformal prediction set replaces a bare softmax score — a low-confidence call is flagged, not silently trusted.

(CSP+SE+BiFPN detector · Swin classifier · Split Conformal Prediction)

Carbon Number, Chosen by Evidence Not Best Score

When two models' bootstrapped error intervals overlap, the more interpretable model wins over a marginally-lower-error black box.

(ElasticNet/MARS vs RF/XGBoost/ANN · 100x bootstrap)

Prove It, Without Exposing It

An auditor verifies a heat's carbon claim on-chain via zero-knowledge proof, without ever seeing the mill's private furnace data or imagery.

(Hyperledger Fabric · on-chain hashes · ZK verification)

INNOVATION & UNIQUENESS

- **Evidence-Linked Grading**
- **Statistically-Honest Model Choice**
- **Blockchain-Backed DPP**
- **Zero-Knowledge Verification**

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Technical Approach

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Computer-Vision Subsystem

- **Detector (CSP+SE+BiFPN):** framed as object detection, not classification, because scrap frames mix multiple contamination types (galvanized sheet, paint, grease, inclusions) at different scales — BiFPN fuses multi-scale features so large sheet scrap and tiny inclusions are caught in the same pass
- **Fine-grained classifier:** Swin Transformer on low-confidence crops — chosen over ResNet/ViT for CNN-like local texture sensitivity (rust, coating residue) plus transformer-level long-range shape context, without ViT's need for huge data or pretraining
- **Uncertainty:** Split Conformal Prediction outputs calibrated prediction sets, not bare softmax — so an industrial grading decision reflects real confidence
- **Explainability:** Score-CAM overlay shows the exact region driving each call, for every classification

Dataset Architecture

- 2,003 Roboflow images, 8 real EFR/BIR scrap grade classes, EDA-verified
- Median bbox coverage 0.998 of frame (85% of images >90% coverage) → annotations are effectively one label per image, not per-object boxes — this finding directly shaped the CV design
- Classification-first components (Swin classifier, conformal prediction, Score-CAM) are validated against this dataset now; the full per-object detector (CSP+SE+BiFPN) is scoped for a separately-collected, manually-boxed set
- Severe 72:1 class imbalance (504 images max class, 7 min class) handled with class-weighted loss + stratified splits, not naive random splitting

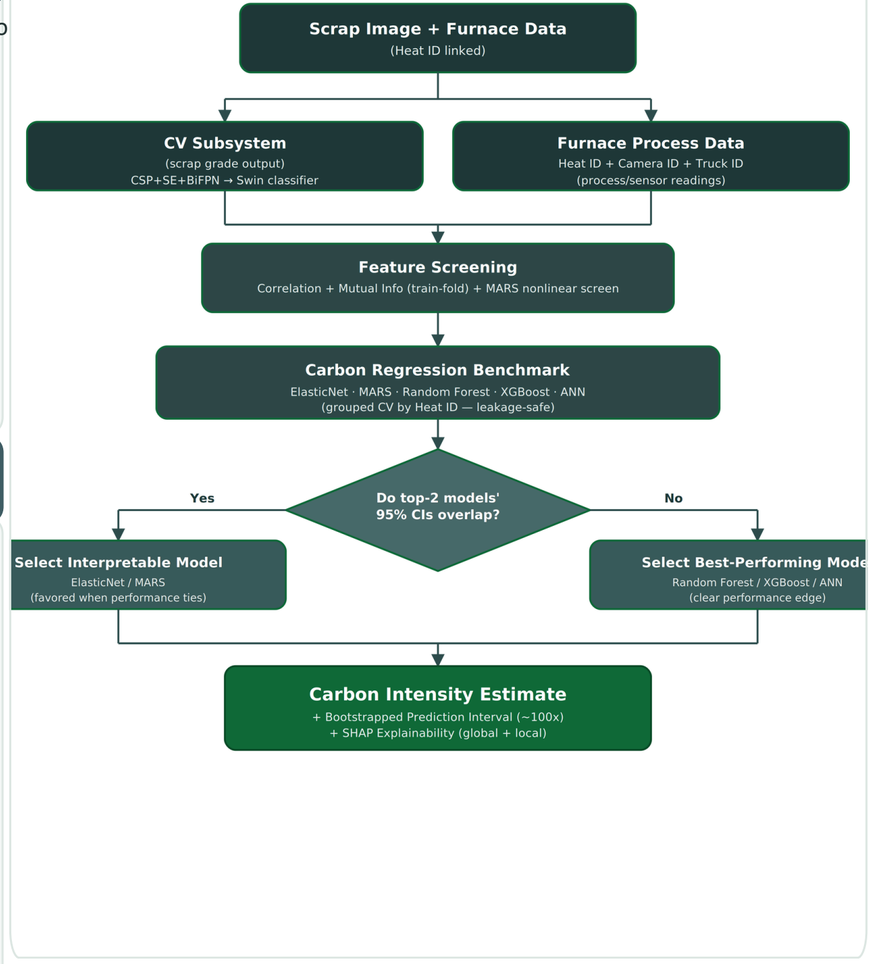
Furnace-Data Fusion & Regression

- **Leakage-safe linkage:** Heat ID + Camera ID + Truck ID join image evidence to furnace records — every evaluation split, not just calibration, is grouped by these keys
- **Feature screening:** correlation + mutual information (train-fold only) plus a MARS-style nonlinear term to catch thresholds a linear screen would miss
- **Benchmark:** ElasticNet, MARS, Random Forest, XGBoost, and a small ANN compared under identical grouped CV + grid search
- **Model selection rule:** when the top two models' bootstrapped 95% CIs overlap, the more interpretable model (ElasticNet/MARS) wins over a marginally-lower-error black box — because a regulator may ask why a heat's number came out the way it did
- **Uncertainty vs. explainability kept distinct:** bootstrapped prediction intervals (~100x resampling) quantify confidence; SHAP (global + local) attributes cause

certificate /audit layer

- **Every field tagged by evidentiary status:** measured / CV-derived / model-derived / provenance — an auditor spots a sensor reading vs. a model estimate at a glance
- **Forms the per-heat DPP record:** Heat ID, steel grade, scrap info, carbon intensity, model version, evidence references
- **On-chain:** only Heat ID, DPP events, content hashes of the evidence, model version, verification status, ZK proof — tamper-evident, not the raw data itself
- **Off-chain:** images, furnace logs, certificates, lab reports — kept off the shared ledger since none of it needs replicating across nodes, and some is confidentiality-sensitive
- **Hyperledger Fabric (**permissioned — known participants: mills, auditors, regulators) with smart contracts controlling who can write a DPP event
- **ZK proof** lets a customer verify a claim (e.g. "carbon intensity below X") without seeing the underlying plant data

Green Steel Passport — Carbon Intensity Pipeline



Feasibility & Viability

Comparison: Why These Model Choices, and Not the Alternatives?

Decision	Our Choice	Alternatives Considered	Why Ours Wins
Fine-grained classifier	**Swin Transformer**	ResNet-50, plain ViT	Shifted-window attention gives CNN-like local inductive bias (surface texture — rust, coating residue) plus transformer-level long-range context (whole-piece shape/size), at near-CNN compute cost. Plain ViT lacks that spatial inductive bias, so it needs heavy pretraining or far more data to match CNN accuracy on a small, specialized dataset — a known published limitation, not project-specific.
Detector attention module	**SE (in CSP+SE +BiFPN)**	CBAM, Coordinate Attention	Published ablations on this task family show CSP+SE+BiFPN beating CSP alone and these alternative attention modules. SE's channel re-weighting is the simplest addition and the best-performing single one for scrap shape/contamination detection specifically — not a claim about attention mechanisms in general.
Carbon regression model	**ElasticNet / MARS (when tied)**	Random Forest, XGBoost, ANN	Selected only when the top two models' bootstrapped 95% CIs overlap, avoiding statistically fragile "wins." ElasticNet/MARS also provide transparent, auditable coefficients—stronger than SHAP alone for regulatory scrutiny. If CIs don't overlap, XGBoost/RF/ANN is used with SHAP explainability.
Ledger	**Hyperledger Fabric**	Public / anonymous chains	mills, auditors, certification bodies, regulators — are known organizations, not an anonymous public. Fabric's identity- and channel-based access control matches that directly; smart contracts enforce who may write a DPP event, so a mill can't unilaterally rewrite its own carbon history.
Classifier uncertainty	**Split Conformal Prediction**	Bare softmax probability	Outputs a calibrated prediction set, not a raw probability — so an industrial grading decision reflects genuine confidence rather than a model's potentially overconfident softmax score.
Regression uncertainty	**Bootstrapped prediction intervals (~100x resampling)**	Single point estimate	Quantifies real confidence around the carbon estimate — kept deliberately distinct from SHAP (which explains cause, not confidence) — and is the actual mechanism the tied-model selection rule in Row 3 depends on.

Technical Feasibility

Every component is a benchmarked model or established technique — detection, transformer classification, conformal prediction, gradient boosting, SHAP, permissioned blockchain, ZK proofs. Integration risk sits in engineering, not unproven research.

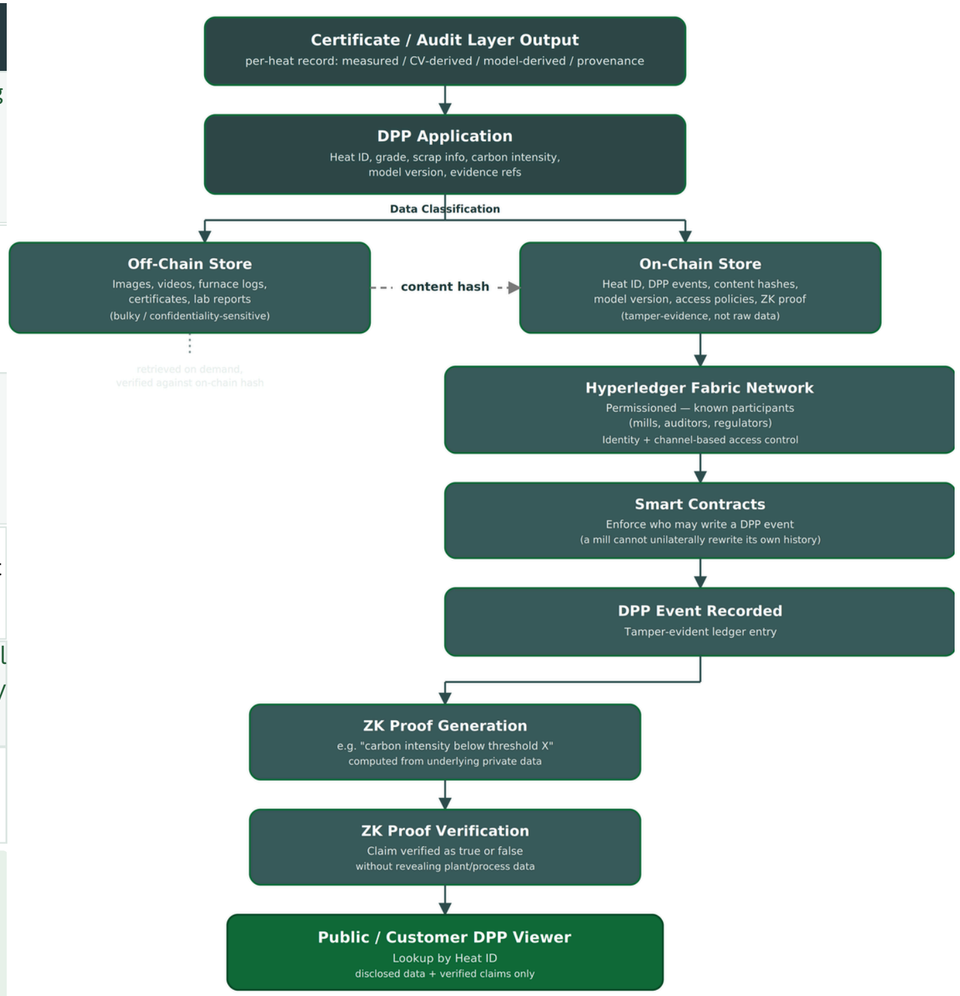
Economic Feasibility

Roboflow scrap images, Kaggle EAF process tables, and open-source ML/blockchain tooling keep a pilot's data and compute cost low against what CBAM non-compliance or a lost green-steel premium would cost an MSME.

Operational Feasibility

The CV, process-fusion, and certificate/DPP subsystems are joined only by Heat ID — any one (detector, regression model, or ledger) can be swapped or upgraded without touching the others.

Green Steel Passport — Blockchain / DPP Verification Flow



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Impact & Benefits

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STAKEHOLDER IMPACT & BENEFITS

MSME Steel Mills (EAF/IF)

ISmall and mid-size induction/EAF mills in India's secondary steel sector typically can't afford dedicated quality-control staff to grade incoming scrap. The CV subsystem gives them that grading for free, consistently, on every batch — not just spot-checks. The payoff is direct: green-steel premiums (from CBAM-compliant buyers, green-financing programs) require documented carbon data, and until now that documentation cost was exactly what priced small mills out.

EU Importers / CBAM Auditors

From January 2026 CBAM's transitional reporting requirements start tightening toward full certificate obligations. Indian exporters currently have no standardized way to produce the embedded-carbon documentation CBAM will require at the border. This system produces exactly that per-heat record, which is what turns a customs clearance from a manual, back-and-forth documentation request into a lookup.

Ministry of Steel / BIS

Right now there's no granular, heat-level visibility into carbon intensity across India's fragmented secondary steel sector — only mill-level self-reporting, if that. This system produces a bottom-up, auditable dataset the ministry doesn't currently have, which is the actual prerequisite for designing any green-steel policy, incentive scheme, or emissions benchmark for this sector. (That's why "Data foundation for green-steel policy" is the one worth pausing on if a judge asks "why does government care" — you're not offering compliance, you're offering the measurement infrastructure that has to exist before policy can.)

Certification Bodies

Certifying a mill's carbon claims today means manual auditor visits and paper trails. An evidence-linked certificate (Score-CAM images, SHAP attributions, hash-verifiable provenance) gives a certifier something to check against rather than something to take on faith, cutting the manual verification burden per heat.

Green Lenders / Financiers

Green loans and sustainability-linked financing need an objective basis to price risk/premiums on. Self-reported carbon numbers from a mill are exactly what a lender can't underwrite against. A third-party, evidence-linked estimate — with the measured/inferred distinction visible — is closer to something a credit committee can actually act on

Climate Regulators

Same logic as the Ministry of Steel box but at the emissions-tracking layer: a transparent scrap-to-carbon pipeline is what lets a regulator actually track sector-level emissions trends over time, instead of relying on periodic, aggregated, self-reported figures.

VALIDATION & DATA FLOW

Dataset validated

grounds the CV claims in an actual, imperfect, real-world dataset (not a toy set), including naming the 72:1 imbalance rather than hiding it.

Leakage-safe linkage

headed off a specific, sophisticated failure mode (a grouped-CV skeptic's objection) before anyone raises it.

Certificate transparency

ties directly back to the evidentiary-status tagging from your Certificate/Audit Layer slide, so a judge who saw that slide sees the thread continue here.

DPP data flow

Off-chain (images, logs, certificates) ↔ On-chain (Heat ID, hashes, model version, ZK proof) via Hyperledger Fabric

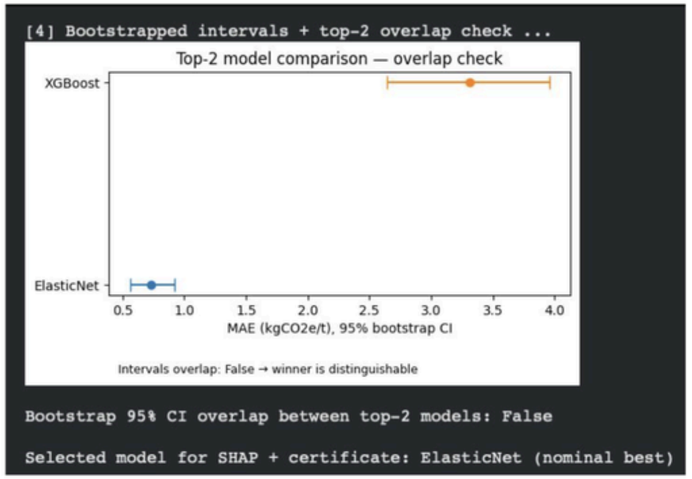
Scope note: the system produces an auditable, evidence-linked estimate — not a certification authority in itself. Every field's evidentiary status is disclosed on the certificate.

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Results: Statistically Rigorous Model Selection

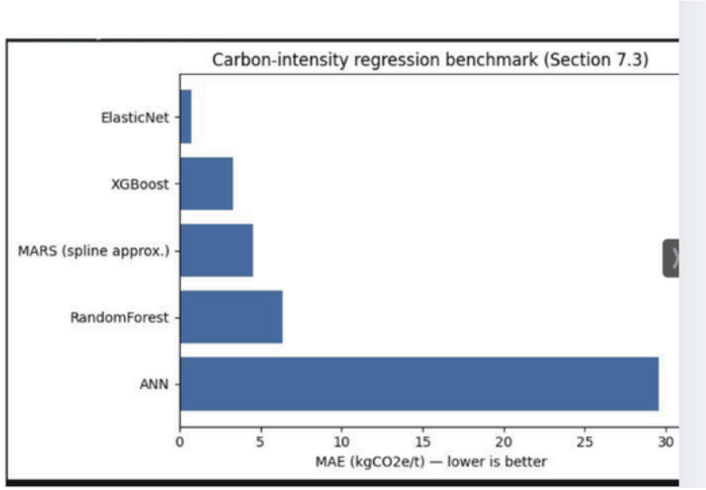
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Top-2 Model Comparison — Bootstrapped 95% CI



ElasticNet vs XGBoost MAE, resampled ~100x.
Intervals do not overlap → ElasticNet is the distinguishable winner.

Carbon-Intensity Regression Benchmark



All 5 candidate models compared under identical
grouped cross-validation (MAE, kgCO2e/t — lower is better).

Full Benchmark Results

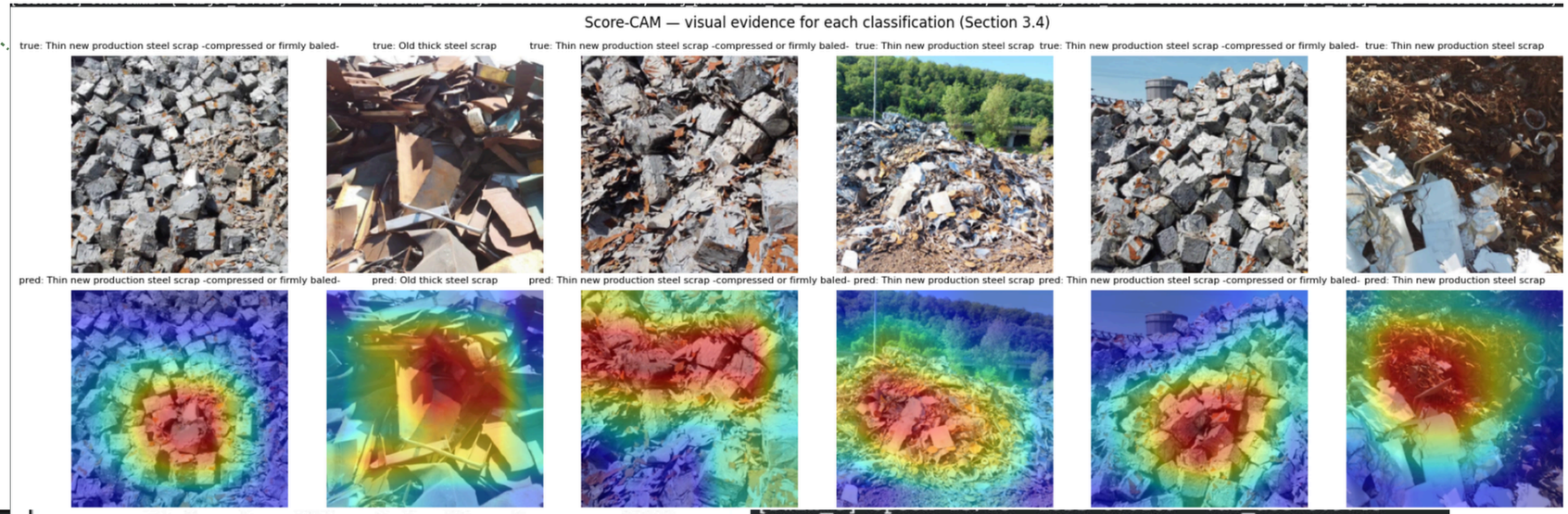
Model	Best Params	MAE	RMSE	R^2
ElasticNet	**alpha=0.001, l1_ratio=0.5**	**0.716**	**1.203**	**0.986**
XGBoost	learning_rate=0.1, max_depth=3, n_estimators=400	3.326	4.410	0.812
MARS	spline approx., n_knots=4, degree=1	4.530	6.131	0.637
Random Forest	max_depth=None, min_samples_leaf=3, n_estimators=200	6.402	8.343	0.327
ANN	small feed-forward net	~29.8	—	—

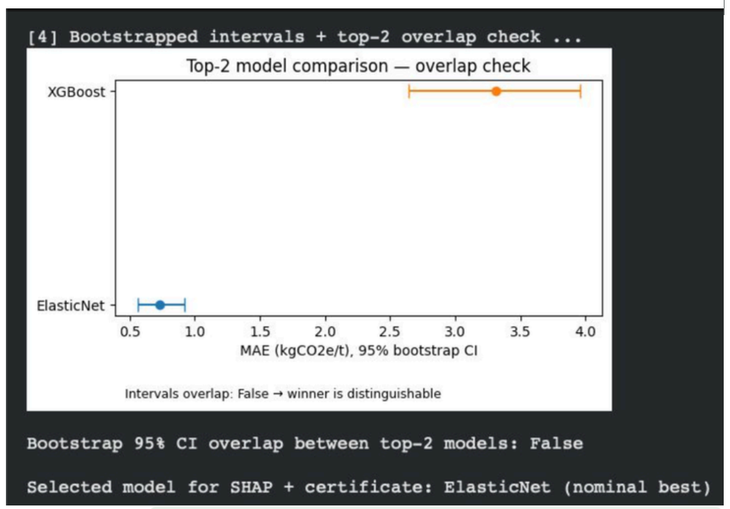
GSP

Results

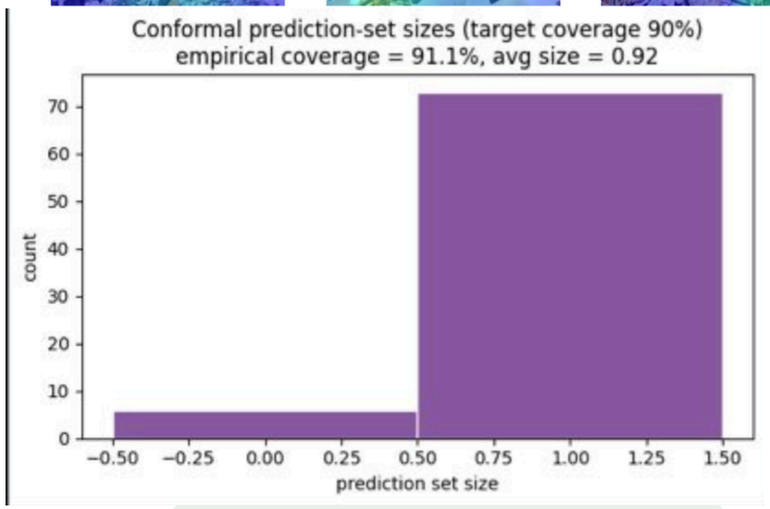
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Score-CAM — visual evidence for every call
Each scrap-grade classification carries a heatmap showing exactly which region of the image drove the decision. Red marks where the model looked hardest. This turns a label into something an auditor can check against the actual scrap pile — not just trust.

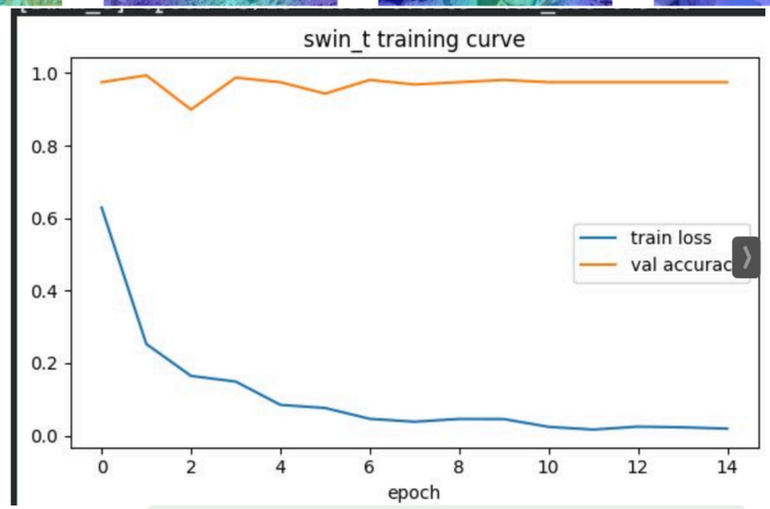




ElasticNet vs. XGBoost, 95% CI from ~100x bootstrap resampling. Intervals don't overlap, so ElasticNet is the statistically distinguishable winner — not just the nominal one.



91.1% empirical coverage against a 90% target, avg. set size 0.92. The model's stated confidence matches its real-world accuracy.



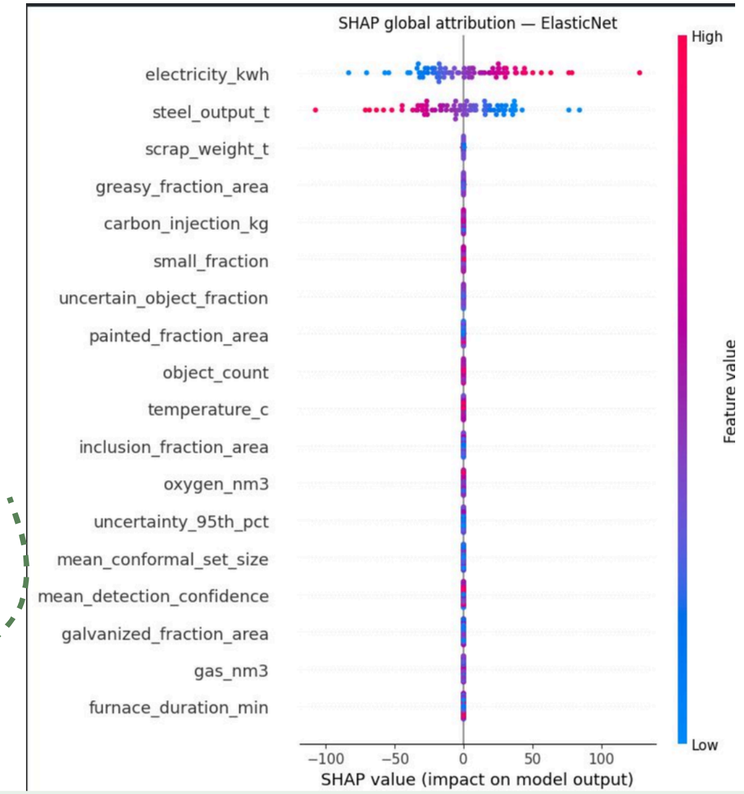
Validation accuracy plateaus near 1.0 while train loss falls smoothly — no overfitting blowout at this scale.

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Results

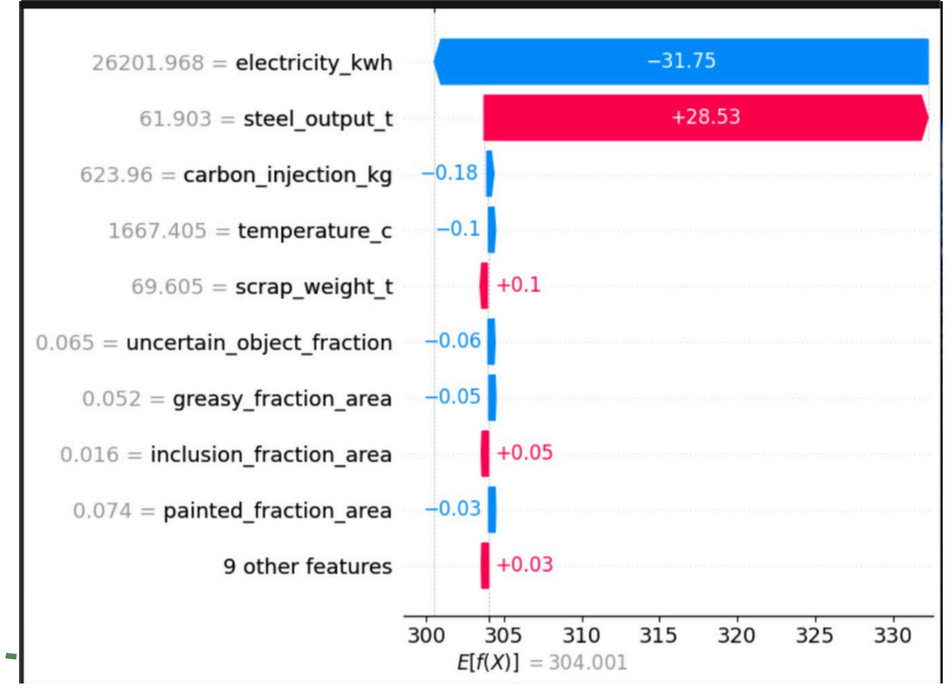
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SHAP global distribution



This is the beeswarm plot: each dot is one heat in the dataset, its position shows how much that feature pushed the prediction up or down, and its color shows whether the feature's value was high (pink/red) or low (blue) for that heat. Read top to bottom, features are ranked by overall importance.
Key read: electricity_kwh and steel_output_t dominate — everything below them (scrap_weight_t, contamination fractions, furnace conditions) contributes comparatively little to the model's predictions. That's a useful finding on its own: it tells you which two process variables actually need tight measurement, and which seventeen don't move the needle much.

SHAP Local explainantion

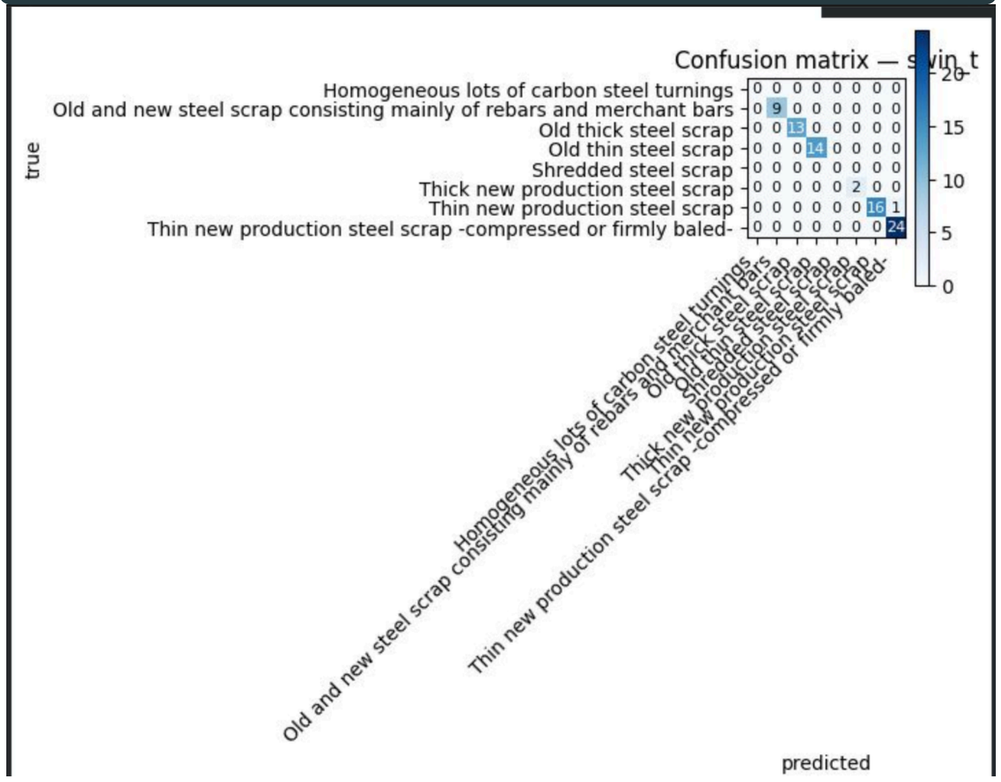


This is a waterfall for one specific prediction: starting from the model's baseline expected output (E[f(X)] = 304.001), each row shows how that heat's actual input value pushed the final estimate up or down, ending at the predicted carbon intensity for this heat.
Key read: for this heat, electricity_kwh = 26,201.968 pulled the estimate down by 31.75, while steel_output_t = 61.903 pushed it up by 28.53 — these two nearly cancel out, with the remaining features contributing smaller adjustments. This is the exact mechanism behind the "model-derived" tag on a certificate: not just a number, but a fully decomposed explanation of how that number was reached.

Results

Metric	Value
Accuracy	0.99
Macro precision	0.745
Macro recall	0.74
Macro F1	0.743
Empirical coverage (target 0.90)	0.911
Avg. prediction-set size	0.924
Singleton sets	92.40%
Empty sets (flagged for review)	7.60%

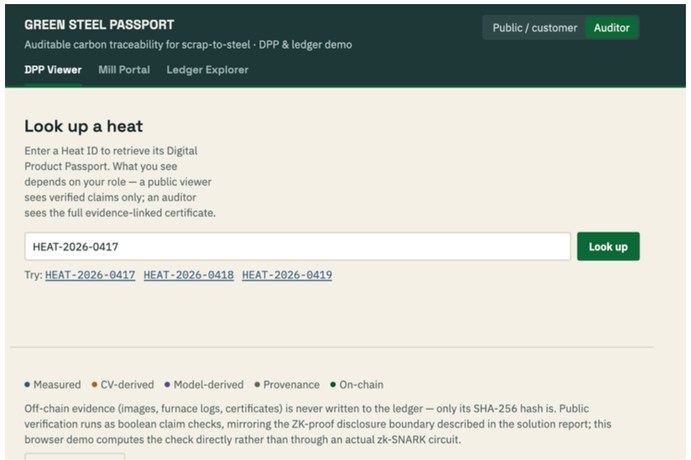
confusion matrix

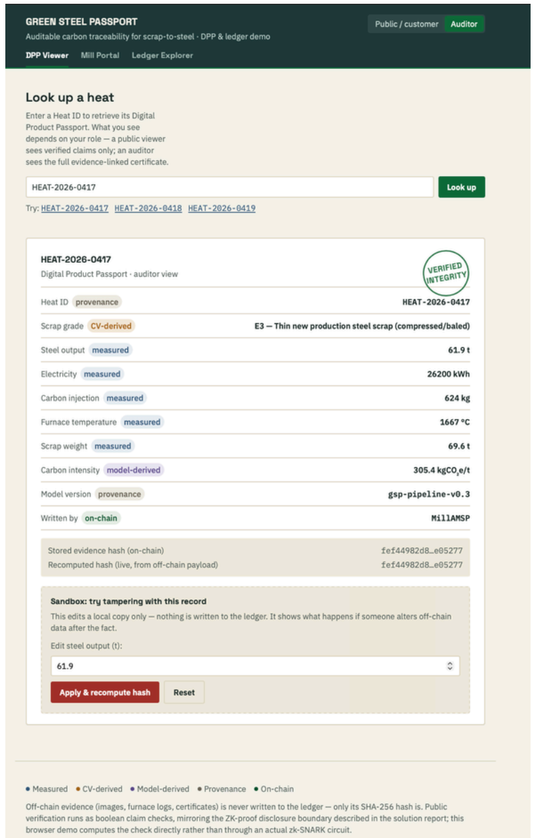


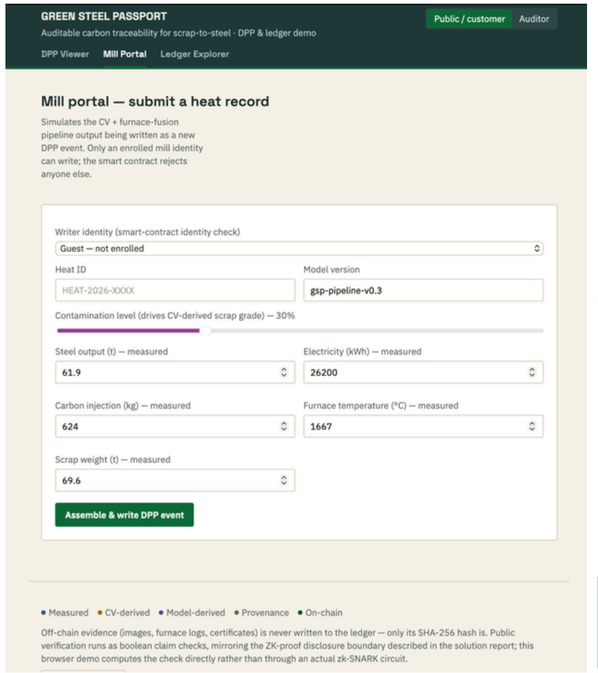
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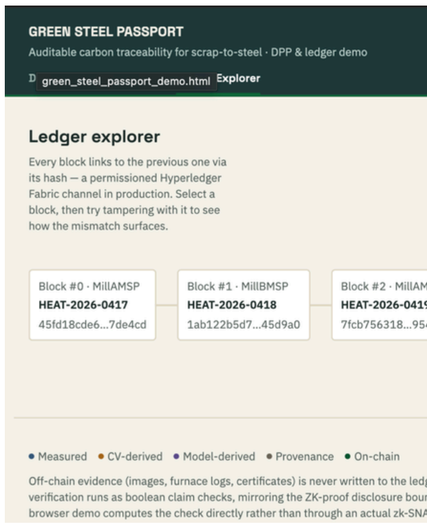
Dashboard

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Research and References

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1. Datasets

- Roboflow scrap-grading export — 2,003 images, 8 real EFR/BIR grade classes, EDA-verified
- Kaggle EAF process tables — gas/O2, materials, chemistry, keyed by Heat ID

2. Computer Vision & Classification

- CSP+SE+BiFPN object detector for multi-object, multi-contamination scrap frames
- Swin Transformer fine-grained classifier, benchmarked against ResNet-50 & ViT baselines
- Split Conformal Prediction for calibrated uncertainty; Score-CAM for visual explainability

3. Carbon-Intensity Regression

- ElasticNet, MARS, Random Forest, XGBoost, small ANN — grouped cross-validation & grid search
- Bootstrapped 95% confidence intervals (~100x resampling) decide the winning model
- SHAP (global & local) attributes each heat's estimate to process and CV-derived features

4. Blockchain, DPP & Verification

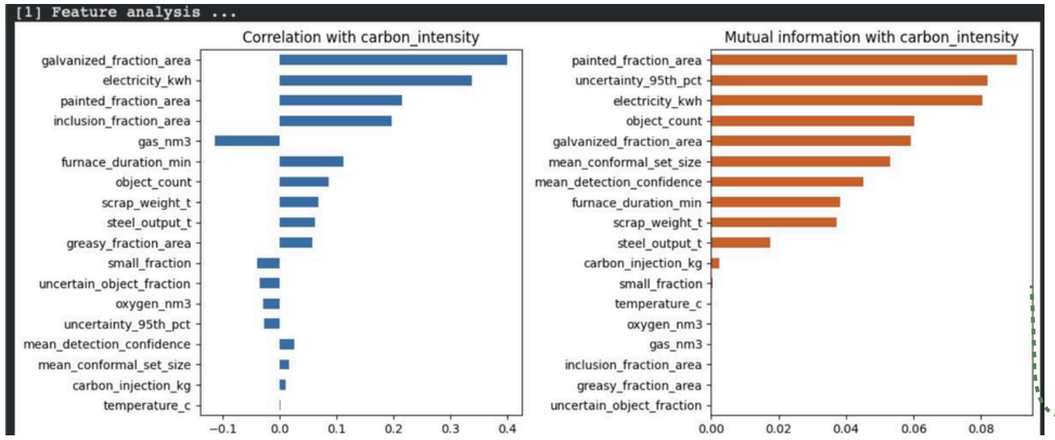
- Hyperledger Fabric — permissioned ledger for known participants (mills, auditors, regulators)
- Digital Product Passport — on-chain hashes, off-chain evidence, one record per heat
- Zero-knowledge proofs for threshold claims without exposing plant-confidential data

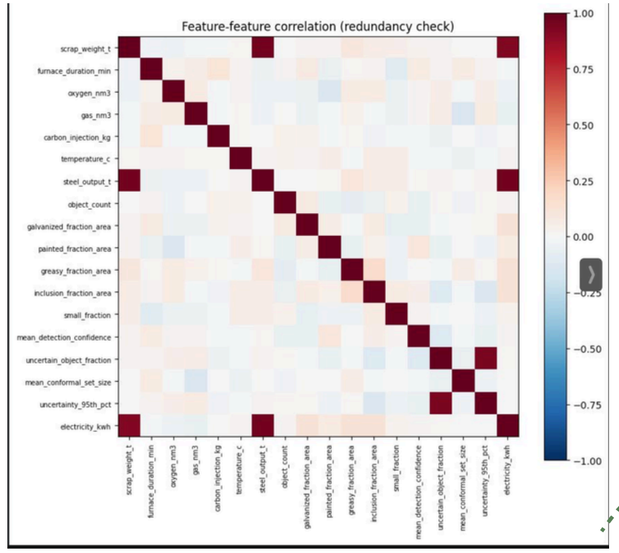
TECH STACK

- CV — Detection + Transformers
- Hyperledger Fabric
- Σ — Gradient Boosting & Stats ML
- ZK — Zero-Knowledge Proofs
- SHAP Explainability
- ID — Heat-ID Data Linkage

GITHUB -->

https://github.com/sahaj200789/buissness-proposal/tree/main





Linear (correlation) and nonlinear (mutual information) screening against carbon_intensity, computed strictly on the train fold. galvanized_fraction_area, electricity_kwh, and painted_fraction_area rank high on both — the two methods agree on what matters, which is itself a useful sanity check

Redundancy check across all 18 candidate features before they enter the regression benchmark. Dark diagonal is each feature against itself; everything off-diagonal stays pale — no pair of features is redundant enough to need dropping.